EXAM PAPERS PRACTICE

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AQA A Level Further Mathematics (7367)

Topic	**B: Complex Numbers**
Sub topic	**B3: Complex Conjugate**

Topic Questions

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AQA A Level Further Mathematics (7367)

Question Paper

Topic	**B: Complex Numbers**
Sub topic	**B3: Complex Conjugate**

To be used by all students and teachers for AQA A Level Further Mathematics (7367) for exams from 2025 onwards.

Students and teachers of other boards may also find this useful

Name : ______________________

Date : ______________________

Time : ______________________

Total Marks Available : _____

Total Marks Received : _____

Q1.

$z = 3 - \mathrm{i}$

Determine the value of zz^*

Circle your answer.

10 $\quad\sqrt{10}\quad$ $10 - 2\mathrm{i}$ $\quad$ $10 + 2\mathrm{i}$

(Total 1 mark)

Q2.

Given that z is a complex number and that z^* is the complex conjugate of z,

prove that $zz^* - |z|^2 = 0$

(Total 3 marks)

Q3.

(a) Solve the equation $w^2 + 6w + 34 = 0$, giving your answers in the form $p + q\mathrm{i}$, where p and q are integers.

(3)

(b) It is given that $z = \mathrm{i}(1 + \mathrm{i})(2 + \mathrm{i})$.

(i) Express z in the form $a + b\mathrm{i}$, where a and b are integers.

(3)

(ii) Find integers m and n such that $z + mz^* = n\mathrm{i}$.

(3)

(Total 9 marks)

Q4.

(a) It is given that $z = x + y\mathrm{i}$, where x and y are real numbers.

(i) Write down, in terms of x and y, an expression for $(z - 2\mathrm{i})^*$.

(1)

(ii) Solve the equation

$$(z - 2\mathrm{i})^* = 4\mathrm{i}z + 3$$

giving your answer in the form $a + b\mathrm{i}$.

(5)

(b) It is given that $p + q\mathrm{i}$, where p and q are real numbers, is a root of the equation $z^2 + 10\mathrm{i}z - 29 = 0$.

Without finding the values of p and q, **state** why $p - q\mathrm{i}$ is **not** a root of the equation

EXAM PAPERS PRACTICE

$z^2 + 10iz - 29 = 0$.

(1)
(Total 7 marks)

Q5.

(a) Solve the following equations, giving each root in the form $a + bi$:

(i) $x^2 + 9 = 0$;

(1)

(ii) $(x + 2)^2 + 9 = 0$.

(1)

(b) (i) Expand $(1 + x)^3$.

(1)

(ii) Express $(1 + 2i)^3$ in the form $a + bi$.

(3)

(iii) Given that $z = 1 + 2i$, find the value of

$$z^* - z^3$$

(2)
(Total 8 marks)

Q6.

It is given that $z = x + iy$, where x and y are real numbers.

(a) Find, in terms of x and y , the real and imaginary parts of

$$i(z + 7) + 3(z^* - i)$$

(3)

(b) Hence find the complex number z such that

$$i(z + 7) + 3(z^* - i) = 0$$

(3)
(Total 6 marks)

Q7.

(a) It is given that $z_1 = \frac{1}{2} - i$.

(i) Calculate the value of z_1^2, giving your answer in the form $a + bi$.

(2)

(ii) Hence verify that z_1 is a root of the equation

$$z^2 + z^* + \frac{1}{4} = 0$$

(2)

(b) Show that $z_2 = \frac{1}{2} + \mathrm{i}$ also satisfies the equation in part (a)(ii).

(2)

(c) Show that the equation in part (a)(ii) has two equal **real** roots.

(2)
(Total 8 marks)

Q8.

It is given that $z = x + \mathrm{i}y$, where x and y are real.

(a) Find, in terms of x and y, the real and imaginary parts of

$$(z - \mathrm{i})(z^* - \mathrm{i})$$

(3)

(b) Given that

$$(z - \mathrm{i})(z^* - \mathrm{i}) = 24 - 8\mathrm{i}$$

find the two possible values of z.

(4)
(Total 7 marks)

Q9.

The complex number z is defined by

$$z = 1 + \mathrm{i}$$

(a) Find the value of z^2, giving your answer in its simplest form.

(2)

(b) Hence show that $z^8 = 16$.

(2)

(c) Show that $(z^*)^2 = -z^2$.

(2)
(Total 6 marks)

Q10.

It is given that $z = x + \mathrm{i}y$, where x and y are real numbers.

(a) Find, in terms of x and y, the real and imaginary parts of

$$(1 - 2\mathrm{i})z - z^*$$

(4)

(b) Hence find the complex number z such that

$$(1 - 2i)z - z^* = 10(2 + i)$$

(2)
(Total 6 marks)

Q11.

The complex number 2 + 3i is a root of the quadratic equation

$$x^2 + bx + c = 0$$

where b and c are real numbers.

(a) Write down the other root of this equation.

(1)

(b) Find the values of b and c.

(4)
(Total 5 marks)

Q12.

The complex number z is defined by

$$z = x + 2i$$

where x is real.

(a) Find, in terms of x, the real and imaginary parts of:

(i) z^2;

(3)

(ii) $z^2 + 2z^*$.

(2)

(b) Show that there is exactly one value of x for which $z^2 + 2z^*$ is real.

(2)
(Total 7 marks)

Q13.

It is given that $z_1 = 2 + i$ and that z_1^* is the complex conjugate of z_1.

Find the real numbers x and y such that

$$x + 3iy = z_1 + 4iz_1^*$$

(Total 4 marks)

Q14.

It is given that $z = x + iy$, where x and y are real numbers.

(a) Find, in terms of x and y, the real and imaginary parts of

$$3iz + 2z^*$$

where z^* is the complex conjugate of z.

(3)

(b) Find the complex number z such that

$$3iz + 2z^* = 7 + 8i$$

(3)

(Total 6 marks)

Q15.

It is given that $z = x + iy$, where x and y are real numbers.

(a) Find, in terms of x and y, the real and imaginary parts of

$z - 3iz^*$

where z^* is the complex conjugate of z.

(3)

(b) Find the complex number z such that

$z - 3iz^* = 16$

(3)

(Total 6 marks)

Q16.

(a) (i) Calculate $(2 + i\sqrt{5})(\sqrt{5} - i)$.

(3)

(ii) Hence verify that $\sqrt{5} - i$ is a root of the equation

$$(2 + i\sqrt{5})\, z = 3z^*$$

where z^* is the conjugate of z.

(2)

(b) The quadratic equation

$x^2 + px + q = 0$

in which the coefficients p and q are real, has a complex root $\sqrt{5} - i$.

(i) Write down the other root of the equation.

(1)

(ii) Find the sum and product of the two roots of the equation.

(3)

(iii) Hence state the values of p and q.

(2)

(Total 11 marks)

Q17.

It is given that $z = x + \mathrm{i}y$, where x and y are real numbers.

(a) Write down, in terms of x and y, an expression for

$$(z + \mathrm{i})^*$$

where $(z + \mathrm{i})^*$ denotes the complex conjugate of $(z + \mathrm{i})$.

(2)

(b) Solve the equation

$$(z + \mathrm{i})^* = 2\mathrm{i}z + \mathrm{I}$$

giving your answer in the form $a + b\mathrm{i}$.

(5)

(Total 7 marks)

Q18.

It is given that $z = x + \mathrm{i}y$, where x and y are real numbers.

(a) Write down, in terms of x and y, an expression for z^*, the complex conjugate of z.

(1)

(b) Find, in terms of x and y, the real and imaginary parts of

$$2z - \mathrm{i}z^*$$

(2)

(c) Find the complex number z such that

$$2z - \mathrm{i}z^* = 3\mathrm{i}$$

(3)

(Total 6 marks)